

PROFESSIONAL DENTAL 3D PRINTING GUIDE

Complete Guide to 3D Printing for Dental Technicians

Nesting, materials, MDR compliance and ROI calculation — from theory to lab practice

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Complete Guide to 3D Printing for Dental Technicians

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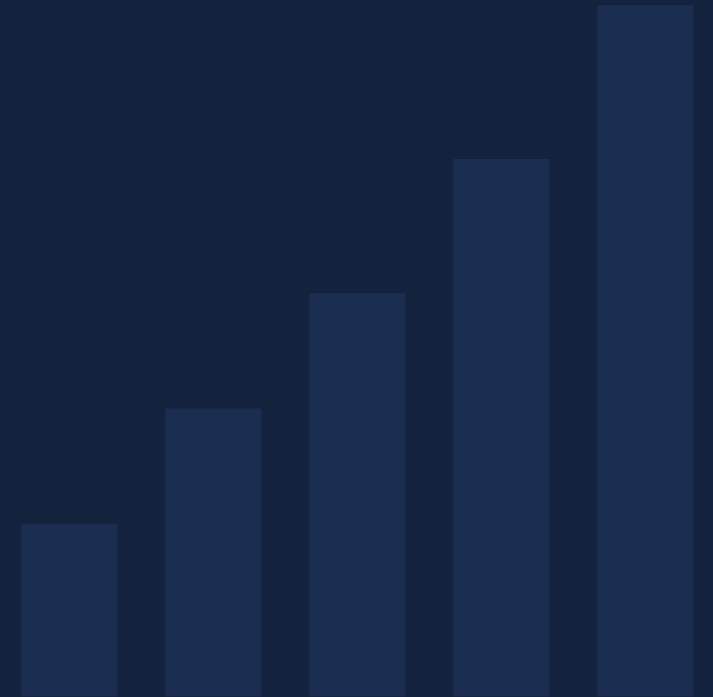
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01

Payback and ROI: Evaluating the 3D Printing Investment

Before signing an order, check how many months it takes to pay for itself



Why calculate payback and ROI before investing

A 3D printer, its consumables and post-processing equipment are a capital investment — evaluating it with the same criteria as any other business purchase avoids surprises in the lab's cash flow.

- The initial investment is not just the printer: it includes the starter package, curing chamber, wash station, and any accessories such as an air extractor or scanner kit
- Cost per unit varies widely with actual volumes: resin, supports/process consumables, energy, labour and depreciation must all be added up case by case
- Comparing technologies (Asiga DLP, Formlabs SLA, Stratasys PolyJet) significantly changes the cost structure: tanks/printheads, consumable lifespan and print speed all vary
- A realistic payback plan must also include a conservative scenario: machine utilisation below 100%, rejects and reprints, and material waste
- Deciding “by eye” is the first mistake: a structured calculation model reveals what would otherwise stay hidden in fixed costs

Payback period, ROI and break-even point: the definitions

Payback period

the time — in months — needed for the operating profit generated by the investment to pay back the net capital invested

ROI (Return on Investment)

the percentage ratio between annual operating profit and the net investment made; it measures how much the investment yields, not how quickly it pays back

Break-even point

the minimum volume of units the lab must produce and sell to cover annual fixed costs, before generating margin

Cost per unit

the sum of material, process consumables, energy, labour and the share of fixed costs (depreciation, service) attributed to each unit produced

Read together, these four metrics tell you whether an investment is worthwhile, how long it takes to pay back, and how much safety margin exists on volumes.

The Payback & ROI simulator: how it works

A free online tool to calculate these numbers for your lab's specific case, in a few minutes and without spreadsheets — bvetuschi.github.io/simulatore-payback-roi-3d

A

Initial investment

Printer + starter package, curing chamber, wash station, accessories

B

2026 Hyper-Depreciation

Tax entity type, tax rate, depreciation period — calculates tax savings and real net cost

C

Service and process consumables

Annual service, IPA / glycerine / filters budget

D

Printer consumables

Adapts to the chosen technology: PolyJet printheads, Asiga DLP tanks, Formlabs SLA resin tank

E

Products, volumes and materials

Up to 7 product types: monthly volume, resin/supports, print hours, labour, selling price

F

Energy and labour

Power consumption, energy cost, operator hourly cost

Source: Dental 3D Printer Payback & ROI Simulator — bvetuschi.github.io/simulatore-payback-roi-3d, Barbara Vetuschi

What the tool gives you

- Total annual fixed costs and fixed cost per machine hour
- Full cost per unit (material + consumables + energy + labour + share of fixed costs) for each product type
- Gross margin and percentage margin per product, and total annual revenue/costs
- Annual ROI and payback in months, calculated in both a normal and a conservative scenario (adjustable machine utilisation, rejects, waste)
- Minimum break-even utilisation: the minimum machine usage volume below which the investment does not pay back
- Every result comes with an (i) icon showing the formula used for the calculation — no number is a black box
- PDF export of the report to share with partners, your accountant or the bank

Source: simulatore-payback-roi-3d — Barbara Vetusch. The tool is for illustrative purposes only: it does not replace professional tax or financial advice.

HOW TO USE IT IN YOUR LAB

bvetuschi.github.io/simulatore-payback-roi-3d

1

Enter the real investment

The dealer's quote and your tax rate, not rough estimates

2

Enter your real monthly volumes

By the product type you actually print — even just conservative estimates

3

Compare the two scenarios

Normal and conservative: if payback stays reasonable even in the cautious scenario, the investment holds up

A solid number is worth more than a dealer's sales promise.

02

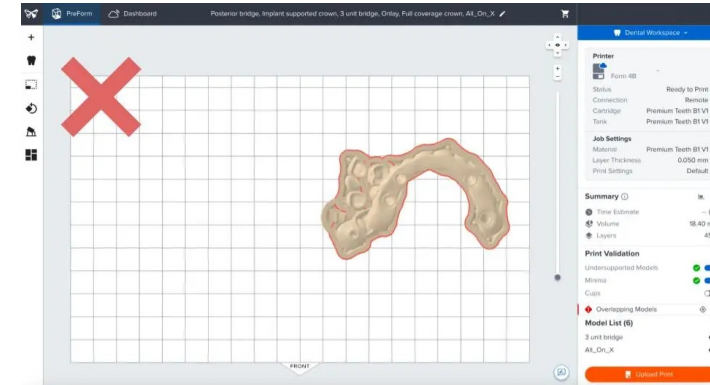
General Nesting & Orientation Principles

The cross-cutting rules that apply to (almost) every indication, from PreForm layout to Asiga Composer supports

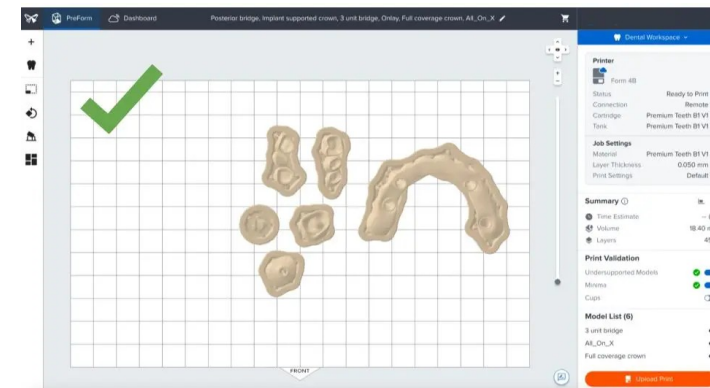


The golden rules of nesting in PreForm

- Always identify the functional surface (occlusal, margins, fit or seating area) and keep it away from support contact points
- Always use a full raft: stability, dimensional accuracy and easier removal from the build platform
- Auto-generate supports, then manually refine critical areas (fit surfaces, connectors, contact points)
- Enable "Printability > Show Minima" for the minimum support density needed — less post-processing
- Layout: no overlap between parts (highlighted in red) — use PreForm's auto-layout
- Always align the insertion axis (die, implant) with the Z axis of the build platform



WRONG — overlapping parts

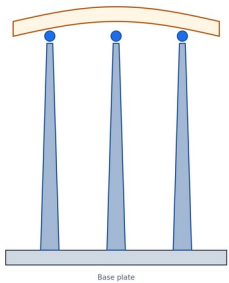


CORRECT — automatic layout

Asiga Composer: the 4 support types

Those working with Asiga printers generate supports in Composer (Ctrl+G), not in PreForm: the logic is the same but the tools have their own names and behaviour.

Column

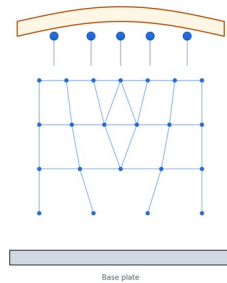


§6.1.6

Rigid, fixed column, tapered — the most common type

When: Practically always: models, bases, technical objects — the default choice

Lattice

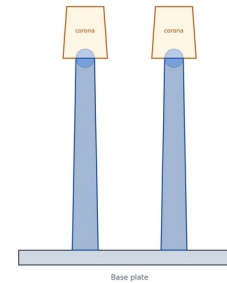


§6.1.3 / §6.5.4

Interconnected mesh of struts; climbs to the model only where needed

When: Complex, delicate pieces where Column would leave too many “pips” to finish

Sprue

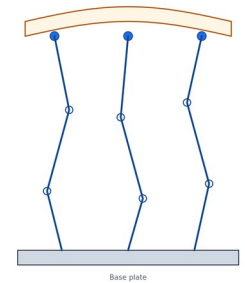


§6.1.7 / §6.5.3

Wide cylinder (a “channel for metal flow”)

When: Practically never in dental technology, at most on the distal ends of bites: designed for metal casting (jewellery)

Flexible



§6.5.1 / §6.1.3

Two joints/hinges between contact point and platform: bends without snapping

When: Aesthetic and delicate surfaces; enabled via preset or from Lattice with Connection = 0

TIP: Auto Support is only the starting point — switch to Manual Editing Mode afterwards to cover unsupported islands or weak margins. Source: Quick Asiga Composer Guide (internal document), from the Composer 2.1 Handbook

Column and Lattice: parameters and selection criteria

RIGID

Column supports

- Geometry: a rigid, tapered column running from the platform (or the model) to a narrow contact point
- Small contact point: leaves a small mark (a “pip”) to be finished with a bur or abrasive paper
- The most stable and predictable of the four: holds up well for heavy pieces and simple geometries
- Default choice for models, bases and technical objects
- Auto-generate, then manually add extra supports on fit surfaces, margins and overhanging areas

FOR DELICATE SURFACES

Lattice supports

- Geometry: a three-dimensional mesh of thin, interconnected struts — introduced in Composer in August 2022
- Each strut climbs progressively toward the piece, distributing the load across multiple points instead of a single column
- Smaller, more distributed final contact point: fewer visible marks to finish on delicate surfaces
- Ideal for complex pieces with many closely-spaced islands (splints, frameworks, detail-dense models)
- Trade-off: a denser mesh means more post-processing time compared to a simple Column

Source: Quick Asiga Composer Guide (internal document), from the Composer 2.1 Handbook §6.1.3, §6.1.6 • Asiga Customer Support Centre (support.asiga.com) — Lattice supports

Sprue and Flexible: the two special cases

ALMOST NEVER IN DENTAL TECHNOLOGY

Sprue supports

- Geometry: a wide cylinder, designed as a casting channel (a “channel for metal flow”) for metal casting in jewellery
- In digital dental technology it should almost always be avoided: it leaves a large contact point that is hard to finish on resin
- Its only remaining use: the distal ends of bites and splints, where a robust anchor is needed and the mark is not aesthetically critical
- If it appears automatically on a visible area, replace it manually with Column or Lattice

FOR AESTHETIC AREAS

Flexible supports

- Geometry: two articulated joints between the contact point and the platform, letting the support bend without snapping when removed
- Reduces the “peel” force transmitted to the piece: ideal on thin aesthetic surfaces (teeth, margins, cusps)
- Enabled via a dedicated preset or by converting a Lattice with the Connection parameter set to 0
- Trade-off: more delicate but less rigid than a Column — needs a visual check before printing

The 4 parameters governing every support type

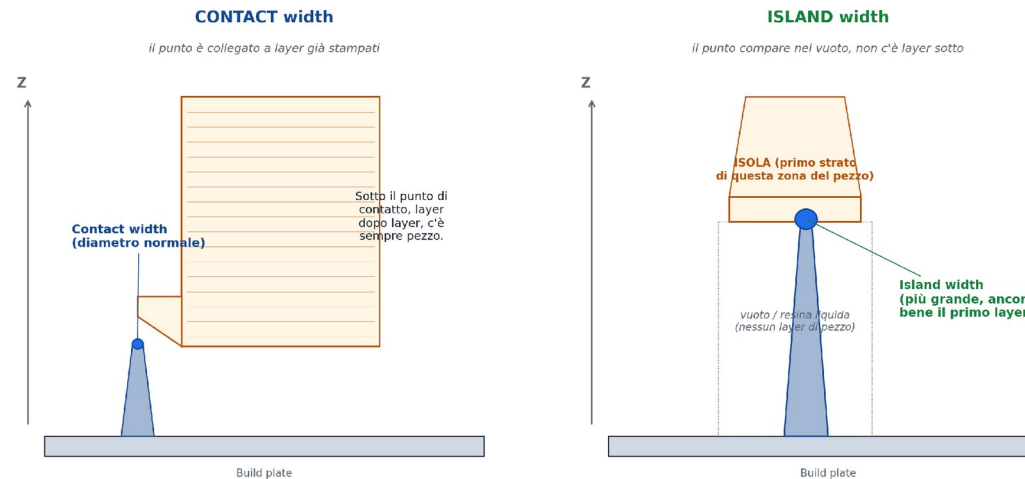
Parameter	What it controls	Practical trade-off
Contact width	Diameter of the point where the support touches the piece	Wider = more stable, but a larger mark to finish
Density (Support spacing)	How many supports are generated per unit of surface area	Denser = less risk of failure, more finishing time
Self-support angle	The tilt beyond which a surface is considered self-supporting (default set in the material's INI file)	A more aggressive angle = fewer supports generated, but risk of unsupported areas
Model Intersupport	Lets a support start from the model's own surface, not only from the platform	Useful for reaching isolated islands; if accidentally left on it creates visible artefacts on the piece

These same four parameters, applied with different values on the same piece, also underpin the dual support generation technique for complex pieces (a “coarse” set + a “fine” set) — see the Bite and Splint section.

Source: Quick Asiga Composer Guide (internal document) — Composer 2.1 Handbook §6.1.5, §8.5, Material INI Definitions

Asiga Composer: Contact width vs Island width

In the Geometry panel two parameters look similar but describe two different moments when a support meets the piece.



- Contact width — where the support touches an area already connected to the rest of the piece (a continuation of the previous layers' geometry). This is the normal case.
- Island width — where the support touches an “island”: an area that, in that layer, appears out of nowhere with nothing underneath to anchor to. This is usually larger than the Contact width, so the island's first layer doesn't detach.

An island is a portion of the piece that first appears in a given layer, not yet connected to the layers below. To reduce islands: re-orient the piece (Auto-Orient → Minimise Islands) or increase Lattice coverage. Source: Composer 2.1 Handbook §6.1.5, §6.1.6, §10.8.3

03

Precision Models

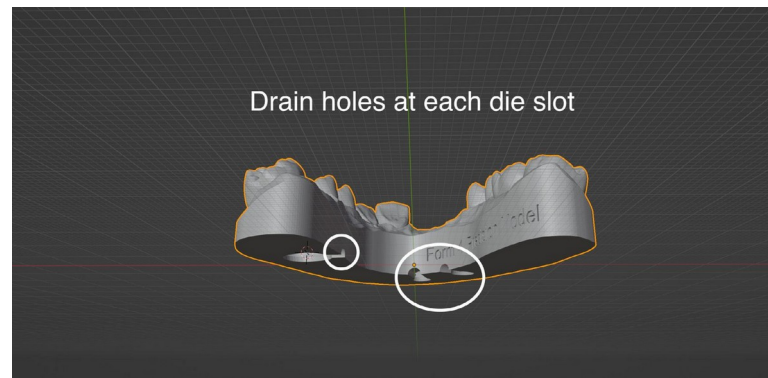
Orientation, drainage holes, supports and die offsets



Direct printing and drainage holes

Print the arches horizontally directly on the platform: the best available support structure, maximum accuracy, no base needed, no overhanging surface

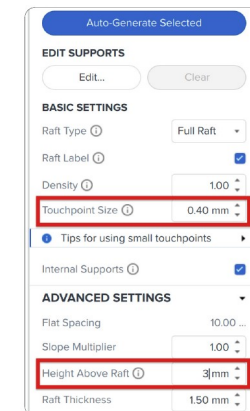
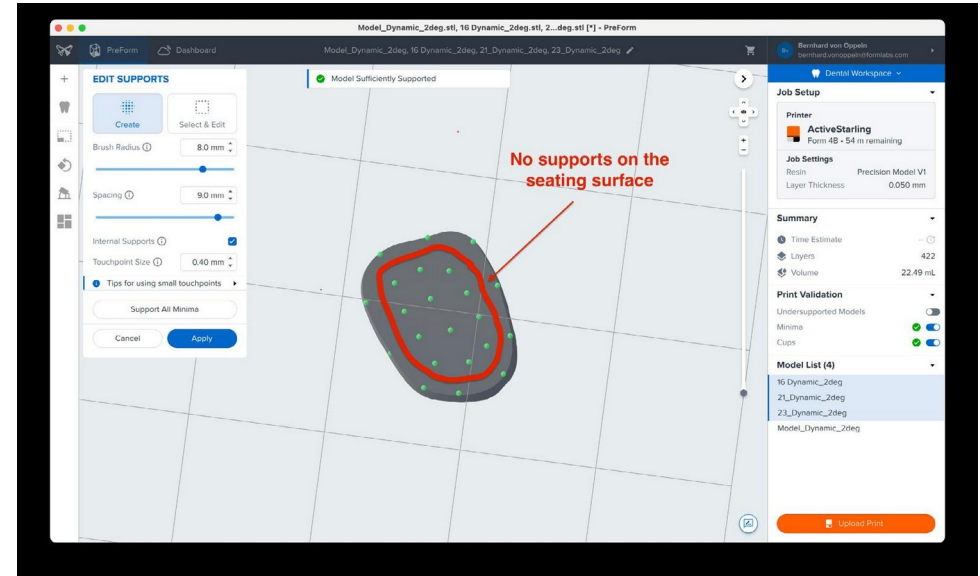
- Drainage holes in every die slot (and in every implant analog socket) prevent the “suction cup” effect from trapped resin during printing
- For implant analog sockets, drainage holes are strongly recommended: they also allow visual inspection of correct seating
- 50 μm layers for maximum precision: 99.9% of surfaces within $\pm 100 \mu\text{m}$, 97.4% within $\pm 50 \mu\text{m}$ (Precision Model Resin on Form 4B)



Source: Formlabs — Precision Model Resin and Form 4B: Guide to Making the Most Accurate Restorative Dental Models

Removable die supports

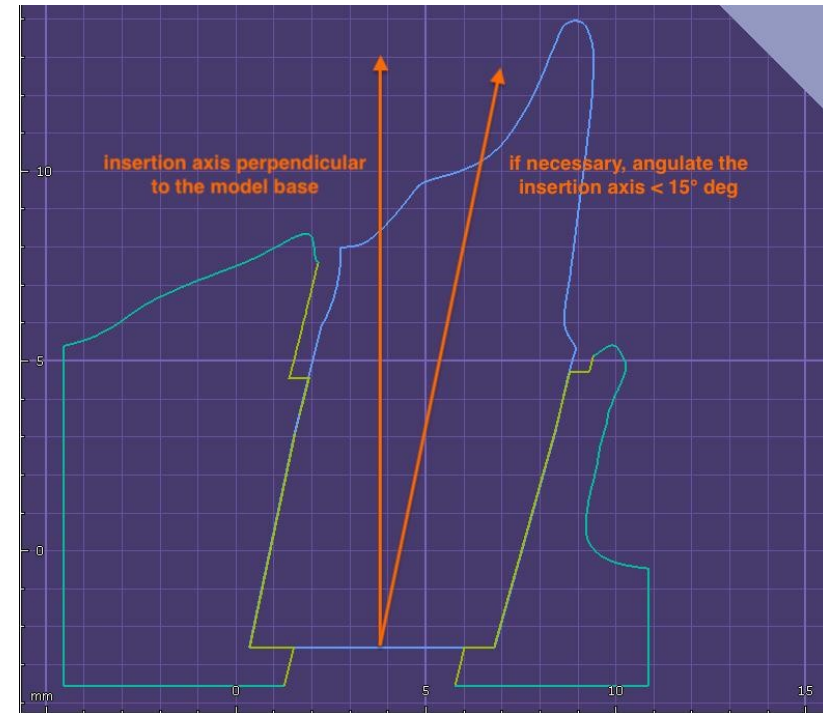
- Generate supports manually for the best results
- Reduce the contact point size to 0.4 mm
- Reduce the height above the base to 3 mm for better stability
- Never place supports on the die's seating area
- Keep supports about 1 mm from the outer edge of the die base



Source: Formlabs — Precision Model Resin and Form 4B Application Guide, p.14

Dies: angulation and offset

Parameter	Recommended value
Model / die offset	0.000 to -0.015 mm
Minimum die slot depth	≥ 15 mm
Maximum recommended angulation	±15° from the Z axis
Taper	2°

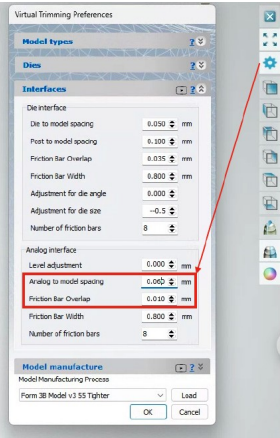


Source: Formlabs — Precision Model Resin and Form 4B Application Guide, p.8 and p.11 (Summary of Removable Die Settings)

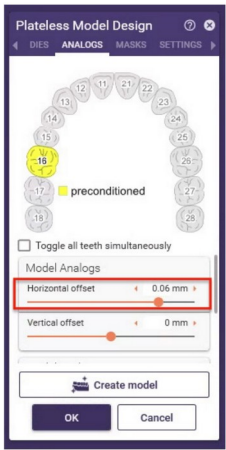
Implant analogs: types and seating offsets

Implant analogs for 3D printed models come in two types, with different seating offsets: the wrong offset compromises the analog's seat in the model.

3Shape Model Builder



exocad Model Creator



Parameter	Recommended value
Analog offset (click-in) — exocad / 3Shape	0.060 mm
Analog offset (screw-retained) — exocad / 3Shape	0.010 mm
Friction bar overlap (3Shape only)	0.010 mm



NT Trading screw-retained analog (left) and Camlog click-in analog (right)

Source: Formlabs — Precision Model Resin and Form 4B Application Guide, p.8-10 (Types of Implant Analogs; Recommended Analog-to-Model Offsets)

04

Occlusal Bites and Splints

Materials, supports and nesting for rigid and flexible bites



Materials and print offsets

RIGID

Dental LT Clear Resin (V2)

- Rigid, fracture-resistant — for rigid occlusal and dental bites
- Offset: 0.05 to 0.10 mm

FLEXIBLE

Dental LT Comfort Resin

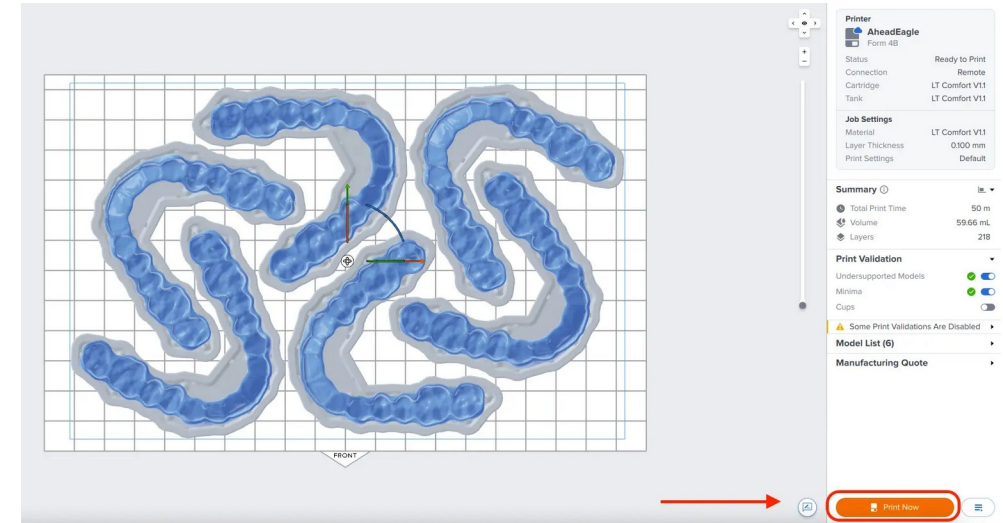
- Flexible and comfortable — bites, night guards, whitening trays
- Offset: 0.01 to -0.03 mm

- In both cases a lower offset value means a tighter fit
- Class II/IIa biocompatible materials, approved for long-term intraoral use in the United States and the European Union
- Always check the minimum occlusal thickness in the chosen material's production guide

Source: Formlabs Dental — Operating manual: 3D printing dental and night guard bites

Supports, contacts and nesting

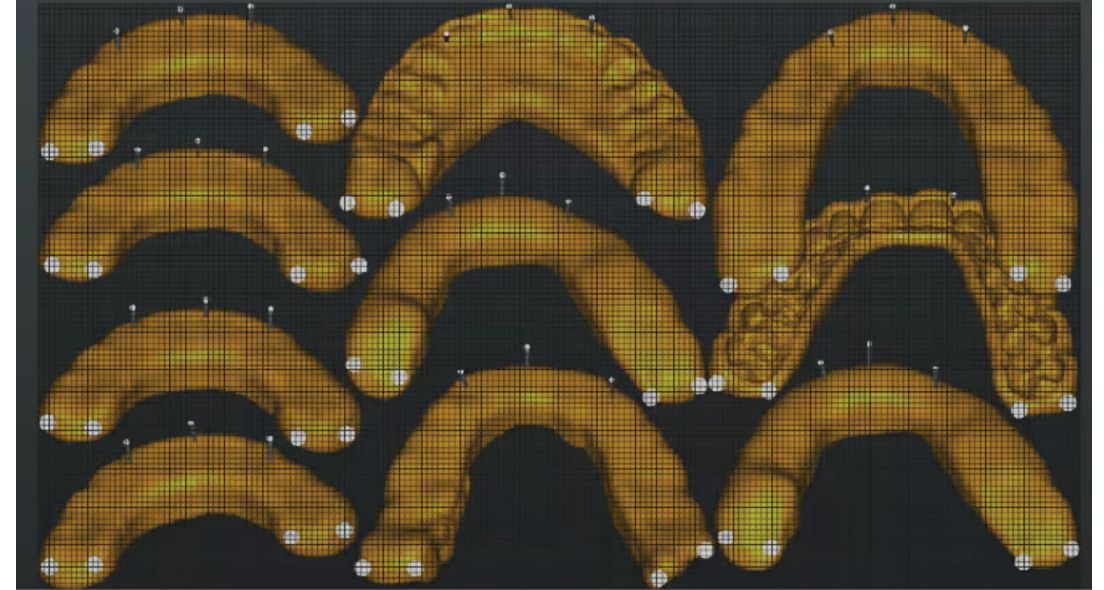
- Auto-generated supports can land in unsuitable areas: opposing-arch contact impressions or dynamic occlusal surfaces — move them manually
- Never reduce the number of supports; always properly support the areas closest to the platform
- Check the green “Printability” icon before starting the print
- For a normal print, position the pieces towards the centre of the platform
- With multiple cases: overlap the rafts to optimise print volume and save material



Source: Formlabs Dental — Bite operating manual, sec. 3.2–3.3 (Adjusting contact points, Layout)

Orientation with Asiga Ultra — tilt and sprue

- Ultragloss / Lift Ultragloss tanks: near-vertical orientation, roughly 75–85° relative to the build platform
- This tilt reduces the contact cross-section per layer and improves resin drainage
- Reduces peel forces per layer, improving occlusal surface quality
- Sprue: 2 supports of 3 mm placed on the sides, at the arch's distal ends
- Sprue removal without clinically relevant marks on the finished bite



Source: Asiga — “An Ultra production upgrade that changes the post-processing equation: UltraGLOSS / LIFT” (asiga.com/blog)

Orientation with Asiga Ultra — central supports and results

- Central supports: 2–3 flexible 1 mm supports in the central anterior area of the arch
- Placed as an overhang, parallel to the build platform, to stabilise the incisor area during printing
- Reduce the risk of deformation in the anterior area, the thinnest and most delicate part of the arch

100 out of 100. Result of Asiga's production test with this configuration: 100 bites printed, 100 accepted — zero print failures, zero detachments.



Source: Asiga — “An Ultra production upgrade that changes the post-processing equation: UltraGLOSS / LIFT” (asiga.com/blog)

Asiga Composer: dual support generation

For complex pieces with many closely-spaced islands — splints, frameworks/partial dentures, models with many islands — a single support set is often not enough. Technique: clone the piece (Ctrl+D+Shift = clone without moving it) and generate two overlapping support sets in the same file, one per clone.

Parameter	“Coarse” set — hold the piece	“Fine” set — prevent deformation
Contact width	2 mm (1 mm for horizontal splints; 2–3 mm for heavy pieces)	0.25–0.30 mm (down to 0.15 mm with some resins)
Self-support angle	5° — more aggressive	INI default value
Density (Support spacing)	Low — only where needed	High — very dense
Model Intersupport (internal supports)	ON	OFF

Model Intersupport = “internal” supports: lets a support start from the model's own surface (not just the platform), connecting the piece to itself to reach islands. On the “coarse” set it should be ON — there are few supports and they need to reach the islands. On the “fine” set it should be OFF: leaving it on is the most common mistake, because it generates supports that start from the surface and leave visible artefacts on the piece.

05

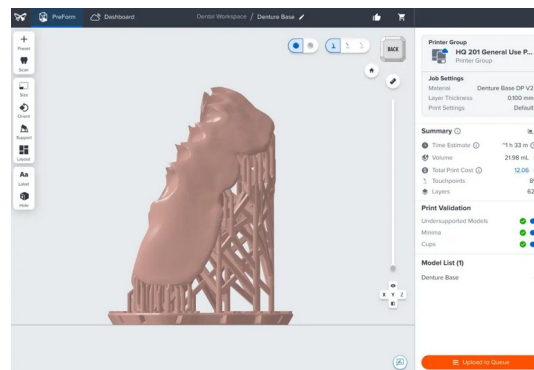
Complete Dentures

Nesting and orientation of denture base and teeth



Nesting: orientation and supports

- Denture base: can be printed at any angle, but beyond 70° layer adhesion may become suboptimal — 60° is recommended for the best accuracy and platform use
- The anterior buccal surface can face either towards the platform or away from it, indifferently
- Place supports on the base's fit (internal) surface, not on the aesthetic one
- Teeth (Premium Teeth Resin): orient the arch at roughly 0–30° relative to the platform; the occlusal surface can face either towards the platform or away from it, depending on the desired finish
- Supports: auto-generate, then manually refine if needed; concentrate contact points on the denture's perimeter and avoid the tooth arches or cervical margin — use cross-bars if extra support is needed



Source: Formlabs Dental — Digital denture operating manual, sec. 3.2 (Orientation and supports)

Flexible partial dentures: nesting and orientation

- Orient the flexible partial denture horizontally on the build platform
- Place supports mainly on the clasps: they are the thinnest areas and most at risk of deformation during printing and post-processing
- Do not reduce support density on the lingual bar and flexing zones — insufficient support in these areas is among the most common causes of denture distortion
- Always check the printability icon before starting the print: an under-supported clasp is hard to fix once printed

Source: Flexible Partial Denture Design Guide — Asiga Italy Ambassador

Flexible partial dentures: 10 design mistakes to avoid

MISTAKES 1-5

- Clasp engaging the tooth too much (> 0.75 mm) → widens and loses retention (2-6 months)
- Clasp junction too low (< 5 mm) → the clasp fractures at the base (3-6 months)
- Clasp too thin (< 1 mm) → deforms or breaks on first insertion
- Wrong clasp type for the clinical situation → compromised aesthetics, denture hard to insert
- Spur clasp with excessive undercut (> 0.5 mm) → excessive force on the abutment tooth or fracture

MISTAKES 6-10

- Missing mandibular lingual coverage → fracture in the canine area (4-6 months)
- Break in the lingual bar, especially in the canine area → fracture almost certain (3-6 months)
- Connector too narrow at the canines (< 6 mm) → fracture in the canine region, the most common failure site (4-8 months)
- Distal extension too thin, Class I/II (< 2 mm) → instability, food accumulation, fatigue failure
- No quality control before production → every mistake above makes it into production

Flexible partial dentures: the 5 critical measurements

Parameter	Value	Technical note
Clasp engagement on the tooth	0.5–0.75 mm	Only 0.5 mm for the spur clasp
Clasp thickness	1.0 mm	Uniform along the entire length
Lingual bar (height)	8–10 mm	Mandatory and continuous in the mandible
Lingual bar (thickness)	1.5 mm	2 mm in the canine area
High-stress zones (canines, extensions)	2.0 mm	Distal extensions, junctions
Clasp junction height	5.0 mm	4 mm only for the split clasp

Source: Flexible Partial Denture Design Guide — Asiga Italy Ambassador

06

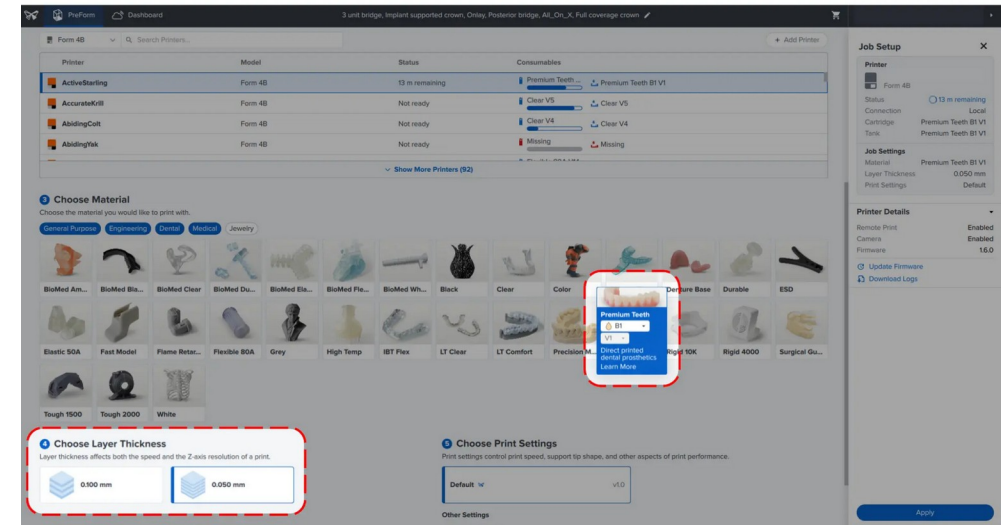
Provisional Restorations

Crowns, bridges and All-on-X immediate loading



Orientation in PreForm

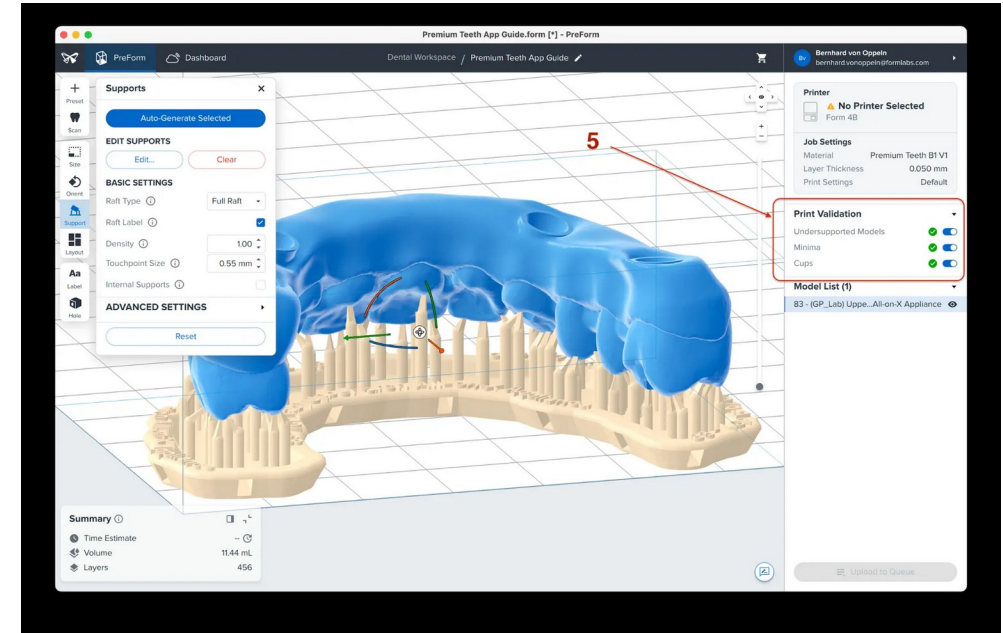
- Full All-on-X arches: angle between 0° and 30°, with the anterior teeth closest to the build platform
- Occlusal surface always facing TOWARDS the platform; fit surface facing outward
- Align implant geometries with the Z axis as closely as possible for optimal accuracy
- Single units and provisional bridges follow the same rule: occlusal towards the platform, fit surface outward
- Print times vary depending on the chosen orientation and job setup



Source: Formlabs Dental — 3D printing of provisional applications, sec. 3.3 (Orientation)

Supports and full raft

- Always use a full raft: accuracy, stability and easier removal from the platform
- Auto-generate supports with “Internal supports” disabled, then manually refine
- Move support points away from fit surfaces, contact points and connector geometries
- Enable “Printability > Show Minima”: it adds only the necessary amount of supports, reducing post-processing
- Default settings are optimised for long-span bridges; reduce density and touchpoints for simpler cases
- Inlays: place supports on the contact surface with the adjacent tooth — it's an area to be finished anyway, so any support marks don't compromise the final result



Source: Formlabs Dental — 3D printing of provisional applications, sec. 3.4 (Support structure generation)

Minimum thicknesses and connectors

Element	Requirement
Single units (crowns, inlays, onlays, veneers)	Minimum wall thickness: 1.0 mm
Bridges up to 7 units — anterior	Wall ≥ 1.0 mm · Connector ≥ 12 mm ²
Bridges up to 7 units — posterior	Wall ≥ 1.5 mm · Connector ≥ 16 mm ² · max 1 pontic molar
Full All-on-X arches	General wall ≥ 1.0 mm · around Ti-base/abutment ≥ 1.0 mm · cement gap per implant manufacturer
All applications	Always disable “Advance shell milling” (exocad) / “Drill Compensation” (3Shape)

Source: Formlabs Dental — 3D printing of provisional applications, sec. 2.1 (Design setting recommendations)

Materials and print parameters

- 50 µm layers for maximum precision — standard for single units, bridges and All-on-X
- 100 µm layers only to speed up printing (immediate loads)
- Premium Teeth Resin: biocompatible nano-ceramic, cementable with traditional provisional cements for up to 12 months
- CAD cement gap: exocad 0.06 mm · 3Shape 0.01 + 0.07 mm extra · 1 mm distance from the margin
- An implant-supported restoration's screw always seats on a metal component (Ti-base) — never directly on the printed resin

Premium Teeth Resin — 4 shades

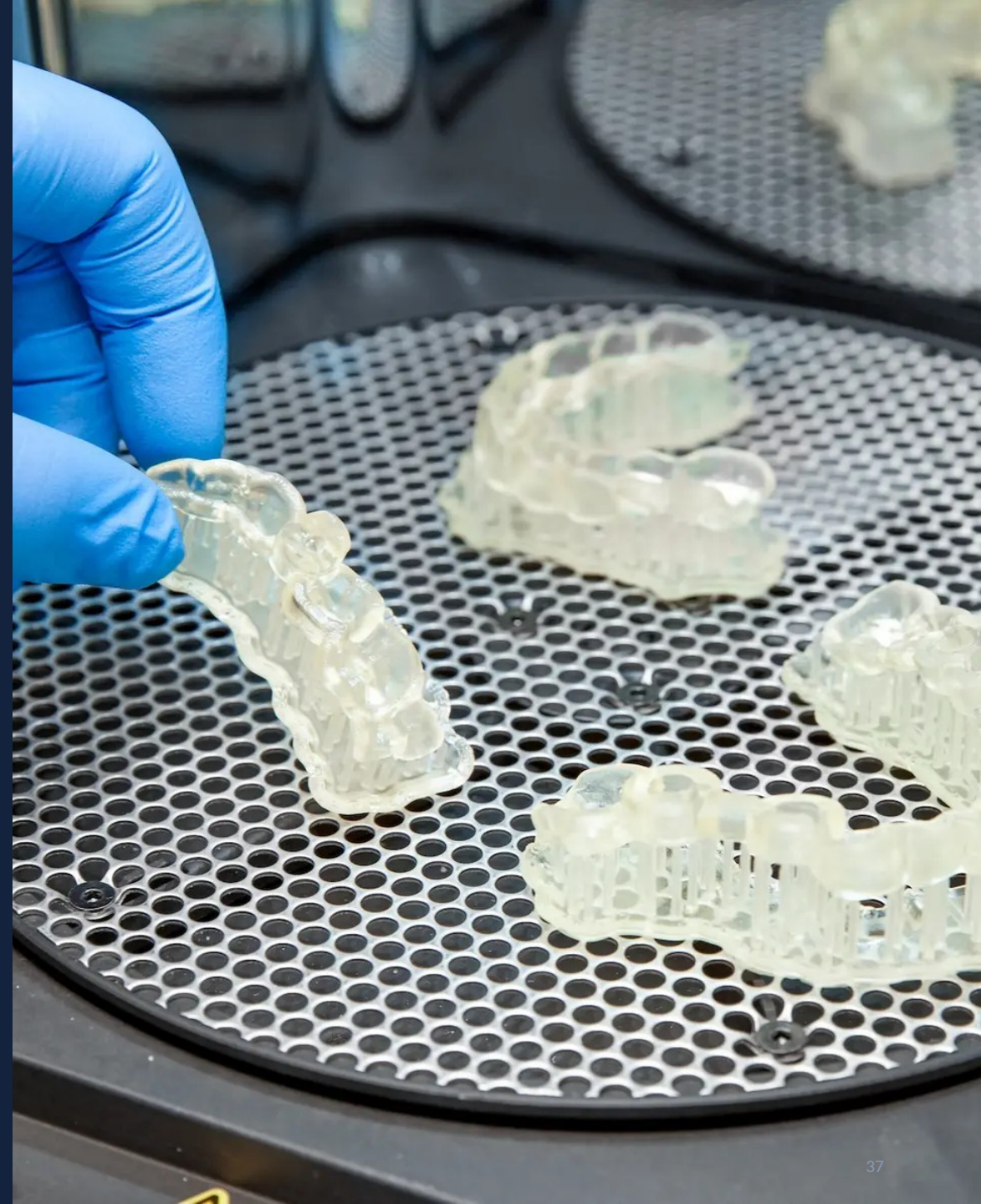


Source: Formlabs Dental — 3D printing of provisional applications, sec. 2.2–3.2

07

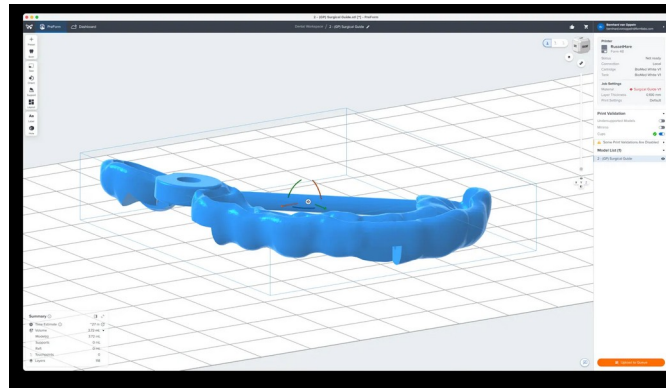
Surgical Guides

Nesting, orientation and supports for implant guides



Orientation in PreForm

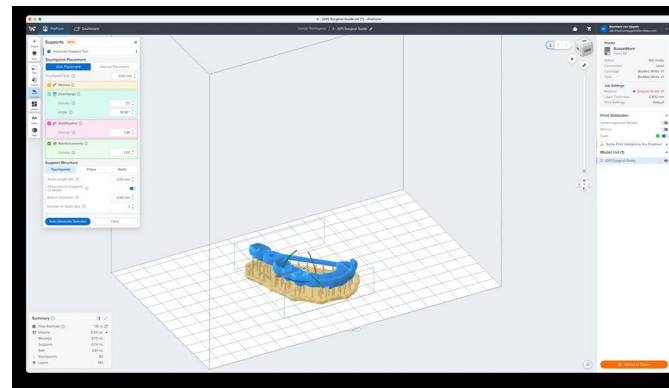
- Orient surgical guides horizontally, with the fit surface facing away from the build platform
- This ensures supports are not generated on the fit surface, the one most critical to assembly accuracy
- If printing several guides in the same session, stagger the position of each model on the platform to optimise print volume
- Always check the alignment of the sleeve holes before starting the print



Source: Formlabs Dental — Surgical guide operating manual, sec. 3.3 (Orienting surgical guides)

Support generation

- Generate supports with the Auto-Generate Selected button
- Make sure there are no supports near the sleeve holes or on the fit surfaces, for simple and accurate assembly
- Use Manual Placement to closely inspect support positions and add or remove them as needed
- Remove supports carefully where they join the guide, to avoid damaging the sleeve holes



Source: Formlabs Dental — Surgical guide operating manual, sec. 3.4 (Generating supports)

Enabling Asiga Transparent Mode

- Transparent Mode is an exposure/projector parameter tied to the printer's firmware — not a display mode
- Improves precision when printing transparent/clear materials; available only on Asiga Ultra and Max 2 — not on Max or Pro 4K
- Enable it when printing guides with different insertion axes (vertical, diagonal, horizontal) on the same platform
- Prevents light from spreading well beyond the layer being cured, reducing lateral overcuring (blooming)
- Slightly increases print time

Source: Asiga Composer 2.1 Handbook, §8.5.6 (Projector Parameters) and Material INI Definitions

08

Material Properties: What to Look for in a Technical Data Sheet

Three numbers to always compare before choosing a resin for clinical use

Flexural strength

Measures the maximum load the material withstands before failing under bending — the first number to check for a restoration that works in occlusion.

- Expressed in MPa (megapascal); measured with a 3-point bending test on a standardised specimen
- Regulatory references: ISO 4049 (polymer-based restorative materials), ISO 10477 (polymer-based crown and veneering materials), ISO 20795-1 (denture base materials)
- ISO 4049 typically requires ≥ 80 MPa for materials intended for load-bearing occlusal surfaces (Type 1) and ≥ 50 MPa for non-occlusal uses (Type 2)
- In dental 3D resins currently on the market, published values typically fall between 85 and 135 MPa, depending on formulation and print technology
- A high value alone is not enough: always check that the test declared in the data sheet matches the clinical indication (provisional vs. definitive, single unit vs. long bridge)

≥ 80 MPa Type 1 — load-bearing occlusal surfaces

≥ 50 MPa Type 2 — non-occlusal use

85–135 MPa Typical range for current dental 3D resins

Source: ISO 4049 — Dentistry: Polymer-based restorative materials · ISO 10477 · ISO 20795-1 · MDPI Prosthesis — “Flexural Properties of Three Novel 3D-Printed Dental Resins Compared to Other Resin-Based Restorative Materials”

Flexural modulus

Measures the material's stiffness — how much it deforms under load. Not to be confused with flexural strength: a material can be strong but flexible, or stiff but brittle.

- Expressed in GPa; the higher the value, the stiffer the material
- Rigid materials (roughly 3–5 GPa for 3D resins, up to 10–15 GPa for reinforced composites or milled PMMA): crowns, bridges, connectors — where chewing loads need to be transferred without bending
- Flexible, low-modulus materials: night guards, splints, trays — where load needs to be absorbed and the material must adapt without fracturing
- Too high a modulus on a long bridge transfers excessive stress to the abutment teeth; too low a modulus on a rigid restoration causes permanent deformation and marginal leakage
- Check that the declared modulus matches the indication: a rigid-bite resin and a flexible-bite resin can have moduli that differ by 5–10 times

Source: comparative literature on the flexural properties of dental resins (3-point bending test) · manufacturer data sheets (e.g. Formlabs Dental)

Fracture toughness

Measures the material's ability to resist the propagation of an existing crack — it predicts how likely chipping is at thin margins or connectors.

- Expressed in $\text{MPa}\cdot\text{m}^{0.5}$; typical tests use pre-notched specimens subjected to controlled bending or tensile loading
- It is often the least reported value in dental resin data sheets compared to strength and modulus — if missing, ask the supplier explicitly for it or use impact strength as a proxy indicator
- Clinically relevant in the highest-risk areas: thin finishing margins, connectors between bridge units, cusps and incisal edges of provisional teeth
- A resin with high flexural strength but low fracture toughness can still chip at thin edges — the three properties must be read together, not in isolation

Source: test methodologies comparable to ISO 13586 / ASTM D5045 (fracture toughness of polymers) • literature on chipping resistance of CAD/CAM and 3D printed materials

How to read a data sheet: comparison checklist

Property	What to ask	Unit	Watch out for
Flexural strength	Does it exceed the ISO threshold for the intended use (occlusal/non-occlusal)?	MPa	Tested dry or after water ageing
Flexural modulus	Is it consistent with rigid (crowns/bridges) or flexible (bites/splints)?	GPa	Compare only materials in the same use category
Fracture toughness	Is it declared? If missing, is there an impact strength figure?	MPa·m ^{0.5}	Thin areas: margins, connectors, cusps
Reference standard	Which ISO/ASTM is cited for the test method?	—	Data are not comparable if the test method differs

Three numbers, read together, are worth more than a marketing slogan: before adopting a new material, compare them against what you already use in the lab.

09

Material Safety: IFU, SDS and MDR Classification

What to check before every clinical use of a new material or device

Checking the IFU (Instructions for Use)

- IFUs are legally binding: they define what can and cannot be done clinically with that material
- Always check: declared provisional or definitive use, contact duration, mechanical limits, specific indication (single unit, bridge, All-on-X)
- A technically valid material isn't automatically covered: check that the intended use is actually declared in the IFU

MDR risk class (e.g. IIa)

- Indicates the device's level of invasiveness. It does NOT say what can or cannot be done clinically: it determines the regulatory pathway, not the clinical indications.

ISO 10993-1 — contact duration

- ≤ 24h: limited contact · 24h–30 days: prolonged contact · > 30 days: long-term contact (cumulative duration, including intermittent use)

Source: Regulation (EU) 2017/745 (MDR) · ISO 10993-1:2018 — Biological evaluation of medical devices, Part 1

Class IIa vs Class IIb — MDR Rule 5

MDR Rule 5 (EU) 2017/745 covers devices that are invasive with respect to body orifices but not surgically invasive: removable, non-implantable, with no tissue contact through the breach of a biological barrier.

CLASS I

Class I

- Short-term use in the oral cavity (surgical guide, impression tray, direct-injection composite guide)

CLASS IIA

Class IIa

- Occlusal bites
- Removable by the patient, even beyond 30 days of cumulative contact with the oral cavity

CLASS IIB

Class IIb

- Implantable device
- Prolonged contact with the CNS or central cardiovascular system
- Intended to chemically modify the body or to release drugs

Key point (Rule 5): for bites, splints and occlusal guards — removable by the patient — it is removability that keeps them in Class IIa beyond 30 days. Crowns, bridges and cemented or screw-retained restorations are NOT removable by the patient and follow a different rule (Rule 8) — see the next slide.

Source: Regulation (EU) 2017/745 (MDR), Annex VIII, Rule 5 — medical device classification rules

Examples of Class IIb devices

DENTAL FIELD

Dental field

- Endosseous dental implants (titanium screws) and implant abutments — surgical anchoring in bone, prolonged bone contact
- Filling materials/cements for permanent use in contact with dentine/pulp in certain indications
- Some oral surgery devices with a long-term structural support function

GENERAL FIELD — TO UNDERSTAND THE RULE'S LOGIC

General field

- Hip or knee prostheses — implantable, long-term bone/joint contact
- Semi-automatic external defibrillators — active devices that deliver energy to the body
- Drug infusion pumps — active devices that continuously administer substances
- Contact lenses for long-term vision correction (> 30 days) — Rule 5, prolonged contact with the ocular surface

Principle common to Rule 8 (implantable and long-term surgically invasive devices): if the device is implantable or remains surgically invasive for more than 30 days, Class IIb applies — or Class III if it involves the heart, central circulatory system or CNS.

Custom-made devices: the alternative pathway to CE marking

Much of what a dental lab produces on prescription — crowns, bridges, dentures, bites — falls under the MDR definition of a “custom-made device”, following a different regulatory pathway from serial CE marking.

- A custom-made device is manufactured according to the written prescription of a qualified healthcare professional, to specific characteristics for a single patient (MDR Art. 2(3))
- The dental lab producing it takes on the role of manufacturer under the MDR — no CE marking is required, but a declaration compliant with Annex XIII is
- The declaration accompanies every custom-made device and includes: manufacturer details, device identification data, conformity with the general safety and performance requirements (Annex I), identification of the intended patient
- For Class III implantable custom-made devices, involvement of a notified body for quality management system assessment is still required (Art. 52(8))
- The technical file must be kept and made available to the competent authorities on request, even though no CE certificate is required

Source: Regulation (EU) 2017/745 (MDR), Art. 2(3), Art. 52(8), Annex XIII — Declaration for custom-made devices

Class IIa vs IIb: what changes in practice for the lab

Aspect	Class IIa	Class IIb
Typical lab examples	Crowns, bridges, partial dentures, bites, orthodontic appliances, implant-supported provisionals	Implant anchoring components (endosseous implant, abutment/T-base) — not the restorations screwed onto them
Custom-made device pathway	Annex XIII declaration, no notified body required	Annex XIII declaration; if not custom-made, notified body involvement for technical documentation
Technical documentation depth	Essential file: materials, process, intended use	More thorough file: more extensive clinical evaluation and risk management
Post-market surveillance	Basic collection of complaints and non-conformities	Reinforced surveillance, more structured reporting

The criterion that shifts a device from IIa to IIb is almost never the lab that produces it, but the component's function — see Rule 8 on the next slide.

Surgical anchorage vs. prosthetic restoration: the difference

Rule 8's criterion is neither removability nor material: it is whether the component surgically penetrates bone tissue, or sits in the oral cavity on an already-present structure.

RULE 8 (BASIC) — CLASS IIB

Surgical anchorage

- Surgically penetrates bone tissue, or remains permanently structurally connected to it (breach of a biological barrier)
- Requires a surgical act to be placed, or is an integral part of an implant system already surgically inserted
- Examples: endosseous implant, mini-implant, osteosynthesis screw, abutment / T-base screwed onto the implant

RULE 8, "PLACED IN THE TEETH" EXCEPTION — CLASS IIA

Prosthetic restoration

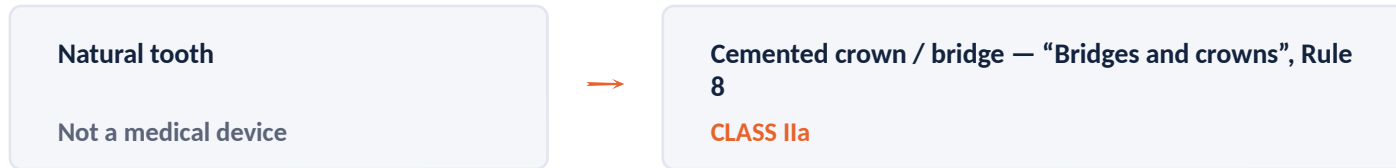
- Does not penetrate bone tissue: it sits in the oral cavity, on a natural tooth or on an already-placed anchorage
- Does not require its own surgical act: it is cemented or screwed onto an already-present structure
- Examples: crown, bridge, inlay, veneer — regardless of the material or fixation technique used

The question to ask: is the component surgically placed in bone (or an integral part of something that is)? If yes → anchorage, Class IIB. If it is simply cemented/screwed in the mouth onto what is already there → restoration, Class IIA.

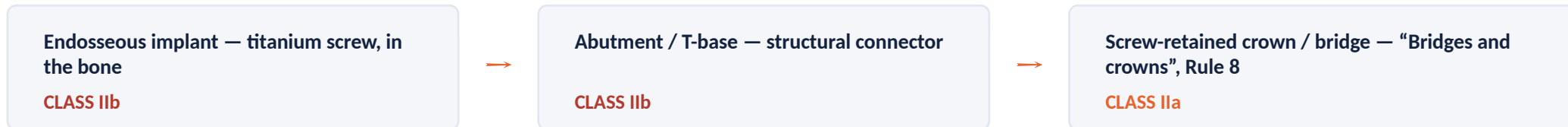
The criterion applied: two practical examples

MDR Rule 8 follows the chain of components, not the clinical case as a whole: each component has its own class.

Scenario 1 — crown/bridge cemented on a natural tooth



Scenario 2 — crown/bridge screwed onto an implant (T-base/abutment)



Key point: the anchorage (implant + abutment) remains Class IIb regardless of the final restoration; the crown or bridge is always assessed as a separate component — Class IIa — whether cemented on a natural tooth or screwed onto an implant. Final classification remains the manufacturer's responsibility, to be verified case by case with the notified body.

Source: Regulation (EU) 2017/745 (MDR), Annex VIII, Rule 8 · MDCG 2021-24 Rev.1 — “Bridges and crowns” example, Class IIa

Checking the SDS — verifying the absence of ACMO and TPO

The SDS (Safety Data Sheet) concerns chemical and operator safety — not clinical use, which is defined in the IFU. Always check Section 3 (composition) and Section 15 (regulatory information) of the material's SDS.

ACRYLOYLMORPHOLINE • CAS No. 5117-12-4

ACMO

- The fastest way to spot it: search for CAS number 5117-12-4 in Section 3 (Composition) of the SDS
- Typical SDS phrases: “may cause damage to organs through prolonged or repeated exposure”, “may cause reproductive effects”
- Substance classified as reprotoxic (fertility risk)

PHOTOINITIATOR

TPO

- A cheap reactive diluent used in some low-cost resins
- Skin sensitiser — inhalation risk from vapours
- The main risk concerns the operator handling uncured resin, not the patient with the finished, properly post-cured device

Best practice: always request the current SDS from the supplier — a missing or incomplete sheet is a sign of poor manufacturer transparency.

Source: Formlabs — “Risk Mitigation: Safety Considerations When Buying a Resin 3D Printer” (formlabs.com/blog)

“Tips & Tricks” Checklist

- ✓ Full raft always, except when printing directly on the platform (models)
- ✓ Functional surface always kept away from support contact points
- ✓ Drainage holes on every die slot and implant analog socket
- ✓ 50 µm layers as the standard for clinical precision
- ✓ “Show Minima” to generate only the supports actually needed
- ✓ No overlap in the layout — always check the green printability icon
- ✓ IFU and SDS: systematic check before adopting a new material
- ✓ Data sheet: flexural strength, flexural modulus and fracture toughness checked before purchase
- ✓ Payback and ROI calculated before every investment in printers or materials

Documents and guides used

- Formlabs Dental — Form 4B Quick Guide: Resources and basic maintenance (internal document)
- Formlabs — Precision Model Resin and Form 4B: Guide to Making the Most Accurate Restorative Dental Models (Application Guide, July 2024)
- Formlabs Dental — Operating manual: 3D printing dental and night guard bites
- Formlabs Dental — 3D printing of provisional applications: full arches for immediate loading, single units and bridges
- Formlabs Dental — Operating manual: 3D printing of digital dentures
- Asiga — “An Ultra production upgrade that changes the post-processing equation: UltraGLOSS / LIFT”
- Quick Asiga Composer Guide (internal document, from the Composer 2.1 Handbook) and support.asiga.com — support types, Contact/Island width, dual support generation, Lattice supports
- Flexible Partial Denture Design Guide — Asiga Italy Ambassador
- Formlabs — Risk Mitigation: Safety Considerations When Buying a Resin 3D Printer (formlabs.com/blog)
- Regulation (EU) 2017/745 (MDR) on medical devices, Annex VIII (Rules 5, 8), Annex XIII, Art. 2(3) and 52(8)
- MDCG 2021-24 Rev.1 — Guidance on classification of medical devices
- ISO 10993-1:2018 — Biological evaluation of medical devices, Part 1
- ISO 4049 — Dentistry: Polymer-based restorative materials
- ISO 10477 — Dentistry: Polymer-based crown and veneering materials
- ISO 20795-1 — Dentistry: Base polymers, Part 1: Denture base polymers
- MDPI Prosthesis — Flexural Properties of Three Novel 3D-Printed Dental Resins Compared to Other Resin-Based Restorative Materials
- Dental 3D Printer Payback & ROI Simulator — bvetuschi.github.io/simulatore-payback-roi-3d, Barbara Vetuschi

Questions?

We're happy to go into more detail on any topic

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